^{36}Ar(pol d,^{3}He) 1993Ma50

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

1993Ma50: a 52-MeV polarized deuteron beam at 15 nA was produced from the Karlsruhe isochronous cyclotron at the Max-Plank-Institut fur Kernphysik. The target was 350-mbar 99.86% enriched ^{36}Ar gas. Reaction products were detected using a Si detector telescope consisting of a 250-μm ΔE-strip detector and a 1.5-mm surface-barrier E detector with FWHM=130 keV. Measured angular distributions of cross sections $\sigma(E(^{3}He),\theta)$ and angular distributions of analyzing powers $iT_{11}(E(^{3}He),\theta)$. Deduced levels, J, π, L-transfers, and spectroscopic factors from DWBA analysis of $\sigma(E(^{3}He),\theta)$ and $iT_{11}(E(^{3}He),\theta)$.

1993Ma50 states that spectroscopic factors were obtained by fitting the forward-angle maxima. The agreement with the analyzing powers is only qualitative. The determination of spins and parities of final states is mainly based on a comparison with measured angular distributions leading to final states with known spins. The comparison with DWBA predictions serves as additional check. The data were taken only in a relatively small angle interval. This is sufficient for the spin determination but not favorable for the measurement of spectroscopic factors.

^{35}Cl Levels

Spectroscopic factor $C^{2}S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=2.95 is a normalization factor used by 1993Ma50 from 1966Ba54.

E(level)[†]	J^{π}[‡]	L[#]	$C^{2}S$[#]	Comments
0	$3/2^{+}$[@]	2	1.72[@]	
1213 *4*	$1/2^{+}$	0	0.78	
1785 *50*	$(5/2^{+})$[&]		0.04[a]	
2676 *7*	$(3/2^{+})$[&]		0.16[a]	
3002 *6*	$5/2^{+}$	2	1.32	
3159 *20*	$7/2^{-}$	3	0.19	
3948 *40*	$(3/2^{+},1/2^{+})$[&]	2	0.03,0.08	E(level): possible doublet of 3918 and 3968 (1990En08) suggested by Table 2 of 1993Ma50.
4839 *12*	$(5/2^{+},1/2^{+})$[&]		0.08,0.03[a]	
5155 *11*	$(5/2^{+})$	2	0.12	J^{π}: $(7/2^{-})$ and $5/2^{+}$ are given in Table 2 of 1993Ma50. Evaluators adopt $(5/2^{+})$, based on DWBA fit of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in Fig. 3 of 1993Ma50.
5583 *13*	$5/2^{+}$	2	0.38	
5720 *11*	$5/2^{+}$	2	1.10	
6136 *8*	$5/2^{+}$	2	0.63	
6745 *12*	$(1/2^{+},5/2^{+})$[&]	(0,2)	0.09,0.28	
6953 *43*	$5/2^{+}$	2	0.49	
7181 *60*	$5/2^{+}$	2	0.20	
7565 *20*	$5/2^{+}$	2	0.30	
8007 *36*	$(1/2^{-},5/2^{+})$[&]	(1,2)	0.16,0.20	
8204 *34*	$5/2^{+}$	2	0.14	
8580 *20*	$5/2^{+}$[&]		0.22[a]	
8921 *70*	$5/2^{+}$[&]		0.18[a]	
9.22×10^{3} *12*	$5/2^{+}$[&]		0.12[a]	
9514 *20*	$(1/2^{-},5/2^{+})$[&]	(1,2)	0.18,0.30	
9.82×10^{3} *12*	$(1/2^{-},5/2^{+})$[&]	(1,2)	0.10,0.26	
1.25×10^{4} *5*				$1d_{5/2}$ is given for a range of 12.0-13.0 MeV in Fig. 4 of 1993Ma50.

† From 1993Ma50.
‡ L+1/2 transfer from analyzing power measurements with $1d_{5/2}$ proton transfer assumed in DWBA calculations, unless otherwise noted.
# From DWBA analysis of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in 1993Ma50. $C^{2}S$ value is extracted for corresponding J^{π} assignment. Where two $C^{2}S$ values are given, they correspond to respective J^{π} values.

Continued on next page (footnotes at end of table)

^{36}Ar(pol d,^{3}He) 1993Ma50 (continued)

^{35}Cl Levels (continued)

[@] L-1/2 transfer from analyzing power measurements with $1d_{3/2}$ proton transfer assumed in DWBA calculations.
[&] Assumed by 1993Ma50 for extracting spectroscopic factors.
[a] No data on $\sigma(E(^3He),\theta)$, $iT_{11}(E(^3He),\theta)$, or L-transfer is given.